# Rust and Vercel: A Visual Introduction

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## At a glance

Rust is a programming language for writing fast, crash-resistant software. Vercel is a cloud platform that puts websites and web apps online in seconds. One is a tool you write code *with*; the other is a place where code *runs*.

| | Rust | Vercel |
|---|---|---|
| What it is | A compiled systems programming language | A cloud platform for deploying web apps |
| Who makes it | The Rust Foundation (started at Mozilla) | Vercel Inc., the company behind Next.js |
| Main promise | Speed like C/C++ with memory safety | From `git push` to a live global site, no servers to manage |
| Typical users | Systems, backend, tooling and infrastructure engineers | Frontend, full-stack and AI app developers |
| Analogy | A precision engine with a strict safety inspector | A delivery service that ships your app worldwide |

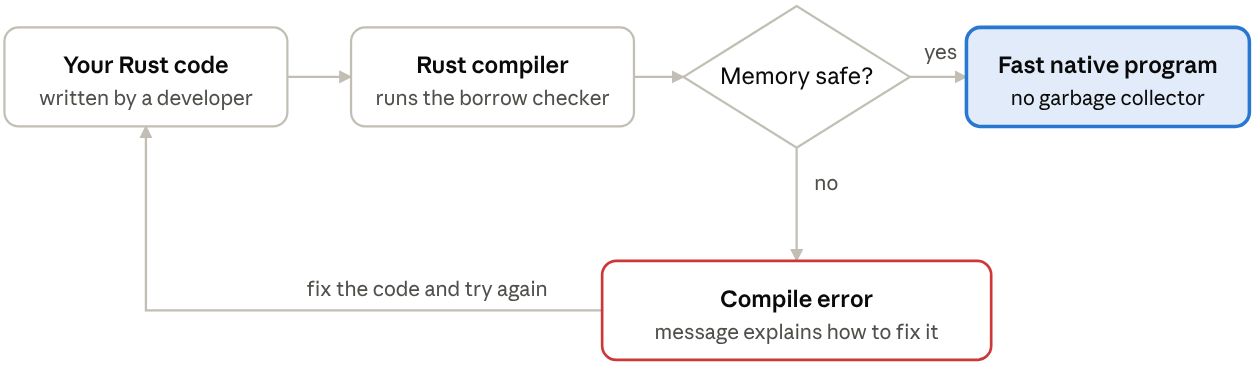
## What is Rust?

Rust is a compiled systems programming language that gives you the speed and low-level control of C and C++ while guaranteeing memory safety without a garbage collector. It was created at Mozilla and is now developed by the Rust Foundation.

The secret is a set of rules called **ownership and borrowing**. Every piece of data has exactly one owner; other parts of the code may *borrow* it, but only under rules the compiler can check. If code could cause a crash or a data race, it simply does not compile.

**The Rust compiler stops unsafe memory use before the program runs**

Ownership and borrowing rules are checked at compile time, not while users run the app




How the Rust compiler keeps programs memory safe

Mistakes such as use-after-free, dangling pointers and data races are caught on the developer's machine, so they never reach users.

## Key features

- **Speed:** compiles to native machine code, comparable to C/C++.
- **Memory safety:** enforced at compile time by the ownership and borrowing model.
- **Strong type system:** enums, pattern matching, traits and generics.
- **Great tooling:** Cargo (build tool and package manager) and crates.io (the package registry), with popular libraries such as tokio, axum, serde and reqwest.

## Where Rust is used

- Rewriting critical components: browsers, search engines, JavaScript runtimes, databases and build tools.
- Networking infrastructure and parts of the Linux kernel.
- Command-line tools, hardware integrations and WebAssembly apps that run in the browser.

## Pros and cons

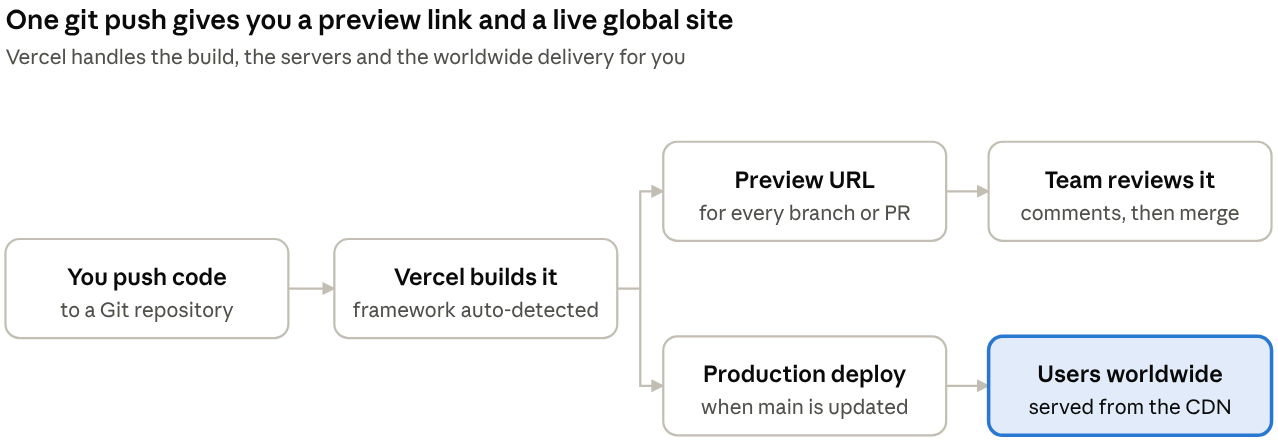
| Pros | Cons |
| --- | --- |
| No memory-related crashes in production | Steeper learning curve than Go or Python |

| Pros | Cons |
| --- | --- |
| Predictable latency, no garbage-collector pauses | Slower to write at first, so higher development cost |
| Very small, fast binaries | Usually not the first choice for a simple web product |

Source: 10Code glossary: Rust

## What is Vercel?

Vercel is a cloud platform that takes the code in your Git repository and turns it into a live website or app, without you managing any servers. It is made by the company behind the Next.js framework, and it describes itself today as an "AI Cloud": developer tools and infrastructure to build, scale and secure web and AI apps.




Vercel's git push to production flow

Every branch gets its own shareable preview link; merging into the main branch publishes the change to everyone.

### Key features

- **Git-based deploys:** connect a repository and every push builds and deploys automatically.
- **Preview deployments:** each branch or pull request gets a unique URL for the team to test.
- **Global CDN:** static files and pages are served from locations close to each user.

- **Vercel Functions on Fluid compute:** backend code runs on demand and scales automatically, billed only for active CPU time.
- **AI tools:** the AI SDK and AI Gateway connect apps to many AI models.
- **Security:** firewall and DDoS protection built in.

### Where Vercel is used

- Marketing sites, online stores and SaaS dashboards, especially those built with Next.js, React, Svelte or Vue.
- AI chat apps and agents that call language models.
- Quick prototypes that need a public link in minutes.

Sources: Vercel AI Cloud, Vercel Ship 2025 recap

## How Rust and Vercel connect

Rust and Vercel meet in two places: Vercel uses Rust to build its own tools, and developers can run Rust code on Vercel.

- **Rust inside Vercel's tools:** Next.js compiles code with SWC and bundles it with Turbopack, both written in Rust for speed. Vercel has also rewritten parts of its Functions infrastructure in Rust.
- **Rust code on Vercel:** the Rust runtime for Vercel Functions is in public beta on all plans. Each Rust file in the `api/` folder becomes a serverless function, built with the `vercel_runtime` crate and run on Fluid compute.

Sources: Rust runtime docs, Rust runtime beta announcement, Vercel Functions powered by Rust

## Glossary

| Term | Meaning |
|---|---|
| Compiler | A program that turns source code into a program the computer can run |
| Garbage collector | A background process that frees unused memory; Rust does not need one |

| Term | Meaning |
| --- | --- |
| Ownership | Rust's rule that each value has exactly one owner at a time |
| Borrow checker | The part of the Rust compiler that enforces ownership and borrowing |
| Cargo | Rust's build tool and package manager |
| Deploy | Publishing code so that people can use it online |
| CDN | A network of servers worldwide that delivers content from a location near each user |
| Serverless function | Backend code that runs on demand; the platform manages the servers |
| Preview deployment | A temporary, shareable copy of a site built from a branch |